

V. Verification Output

(September 28, 2026)

One script re-derives every computational claim in the paper from scratch and reports pass or fail per check. Its current output follows, unedited.

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1. Realisability (thm:realis):  $c^2 = 0$  iff  $F = (s+a)(L_k + t)$ 
[PASS] k=2
[PASS] k=3
[PASS] k=4
[PASS] k=5
[PASS] k=6
[PASS] k=7
[PASS] k=8
2. Antipodal corollary:  $c^2 = 0$  identically for even k, not for odd
[PASS] k=2 (even)
[PASS] k=3 (odd)
[PASS] k=4 (even)
[PASS] k=5 (odd)
[PASS] k=6 (even)
[PASS] k=7 (odd)
[PASS] k=8 (even)
3. k=2 closed form:  $c^2 = -(s_1+s_2)(s_1+s_3)(s_2+s_3)$ 
[PASS] identity
4.  $e_2 = -k$  for every period-one symbol with  $k \geq 3$  (thm:congbound)
[PASS] k=3
[PASS] k=4
[PASS] k=5
[PASS] k=6
[PASS] k=7
[PASS] k=8
[PASS] k=9
[PASS] k=10
5. Integer certificates: the signed adjacency matrix
[PASS] signed Adj P(10,3) nullity 8 --- got 8
[PASS] signed Adj P(20,3) nullity 8 --- got 8
[PASS] signed Adj P(30,3) nullity 8 --- got 8
[PASS] signed Adj P(120,7) nullity 16 --- got 16
[PASS] signed Adj P(30,2) nullity 6 --- got 6
[PASS] signed Adj P(60,2) nullity 6 --- got 6
[PASS] Psi_7 certificate P(7,2) nullity 6 --- got 6
[PASS] Psi_7 certificate P(14,2) nullity 6 --- got 6
[PASS] Psi_7 certificate P(21,2) nullity 6 --- got 6
[PASS] Psi_7 certificate P(28,2) nullity 6 --- got 6
6. k=2 explicit certificate at the three most negative grid values
( $c^2 > 0$  is checked for all n; the nullity is checked two ways, by
counting grid roots of the symbol exactly and, for moderate n only,
by singular values -- the three roots crowd within  $O(1/n^2)$  of -2,
so the matrix is exact but numerically stiff for large n)
[PASS]  $c^2 > 0$  for every  $n \geq 9$ ,  $n \neq 10$  (up to 1001) --- checked 9..199, 500, 1001
[PASS] three prescribed roots are distinct and interior --- checked 9..199, 500, 1001, 5000
[PASS] P(9,2) nullity 6 numerically
[PASS] P(11,2) nullity 6 numerically
[PASS] P(20,2) nullity 6 numerically
[PASS] P(39,2) nullity 6 numerically
[PASS] P(60,2) nullity 6 numerically
7. Saved interval-certified matrices re-evaluated (k=3)
[PASS] P(17,3) nullity 8, edges nonzero --- nullity 8, min edge 0.1462
[PASS] P(18,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2058

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[PASS] P(19,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0145
[PASS] P(20,3) nullity 8, edges nonzero --- nullity 8, min edge 0.1424
[PASS] P(21,3) nullity 8, edges nonzero --- nullity 8, min edge 0.3059
[PASS] P(22,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0878
[PASS] P(23,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0647
[PASS] P(24,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0405
[PASS] P(25,3) nullity 8, edges nonzero --- nullity 8, min edge 0.1584
[PASS] P(26,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0859
[PASS] P(27,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2439
[PASS] P(28,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0296
[PASS] P(29,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2612
[PASS] P(30,3) nullity 8, edges nonzero --- nullity 8, min edge 0.3665
[PASS] P(31,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2290
[PASS] P(32,3) nullity 8, edges nonzero --- nullity 8, min edge 0.0445
[PASS] P(33,3) nullity 8, edges nonzero --- nullity 8, min edge 0.2802
    (17 saved certificates re-evaluated)
[PASS] thm:p17: a saved certificate exists for every n in [17,33] --- all 17 present
8. k=3 identity tiles and their compositions
[PASS] tiles cover a full interval [L,2L] --- lengths 21..43 (23 tiles)
[PASS] single tiles give nullity 8 --- all 23 lengths
[PASS] multi-tile concatenations give nullity 8 --- checked n up to 55
9. Certified tiles: nullity, docking exactness, coverage
[PASS] k=3: every n >= 21 is a sum of certified lengths --- 23 tiles, lengths 21..43 (= [L,2L], so L itself)
[PASS] k=3: every certified tile has nullity exactly 8 --- all 23 tiles
[PASS] k=3: docking weights exact in every certified tile --- all tiles, to 1e-13
[PASS] k=4: every n >= 29 is a sum of certified lengths --- 33 tiles, lengths 29..61 (= [L,2L], so L itself)
[PASS] k=4: every certified tile has nullity exactly 10 --- all 33 tiles
[PASS] k=4: docking weights exact in every certified tile --- all tiles, to 1e-13
[PASS] k=5: every n >= 162 is a sum of certified lengths --- 21 tiles, lengths 54..75; 44 smaller n not covered
[PASS] k=5: every certified tile has nullity exactly 12 --- all 21 tiles
[PASS] k=5: docking weights exact in every certified tile --- all tiles, to 1e-13
10. The paper's own numbers against the computed ones
[PASS] paper's certification count matches the 56 certified tiles --- all say 56
[PASS] Status section names the same k as thm:complete --- both say [2, 3, 4]
[PASS] k=3: paper's L matches the least certified tile length (21) --- L=21
[PASS] k=4: paper's L matches the least certified tile length (29) --- L=29
11. The Krawczyk test itself, re-run in EXACT arithmetic
[PASS] tile 21 at k=3: exact Krawczyk contraction --- alpha = 2.154e-10 < 1 exactly
[PASS] tile 29 at k=4: exact Krawczyk contraction --- alpha = 1.812e-10 < 1 exactly
[PASS] gn 17 at k=3: exact Krawczyk contraction --- alpha = 2.128e-11 < 1 exactly
12. Z values cross-checked against independent solvers
[PASS] Z(P(12,3)) = 7, by every method that completed --- 3/3 completed: reduced 7, unreduced 7, CP-SAT 7
[PASS] Z(P(8,2)) = 5, by every method that completed --- 3/3 completed: reduced 5, unreduced 5, CP-SAT 5
[PASS] Z(P(12,4)) = 6, by every method that completed --- 3/3 completed: reduced 6, unreduced 6, CP-SAT 6
[PASS] recorded cross-validation sweep has no disagreement --- 79 rows agree
14. Reduction on every two-vertex cover (thm:red2)
[PASS] thm:red2: elimination gives the predicted offsets, ends nonzero --- 8 bases, incl. p>1 and w!=0 which thm
[PASS] thm:red2: x -> (x, Lx) is a bijection onto ker A, and null A <= 2(p+q) --- null A = null R, lift residual
[PASS] thm:theta: null A = dim ker(T-I) <= D on theta covers --- 6 bases incl. four-edge; state dimension equals
[PASS] thm:order: order = p+q+max(p,t_m)+max(q,t_m) --- 12 bases, measured vs counted
[PASS] thm:order: order = D iff t_m <= min(p,q) --- the characterisation holds on every base tested
15. Slice lemma (lem:slice)
[PASS] spokes -> +1 and |outer| -> 1, nullity preserved, both parities --- max deviation 2.7e-15
16. Emit the manuscript's numbers from the certificates on disk
[PASS] manuscript numbers emitted from the certificates on disk --- 56 tiles at k=3,4 + 21 at k=5; L(3)=21, L(4)

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ALL CHECKS PASSED

VI. A Certification Transcript

(September 27, 2026)

What “interval-certified” means in practice, for one family of tiles. Each entry is an independent Krawczyk contraction proof with its own unknown count, matrix norm and residual; nothing is shared between them.

certifying 22 identity tiles at $k=3$

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1=22: 398 unknowns, 352 equations, 324 structural, 28 dependent; 40 docking pins, 74 fixed total
      structural 3.11e-61, dependent 2.72e-61, ||Y||=5.39e+02, alpha=6.70e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=23: 419 unknowns, 368 equations, 340 structural, 28 dependent; 40 docking pins, 79 fixed total
      structural 3.11e-61, dependent 4.67e-61, ||Y||=7.55e+02, alpha=9.25e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=24: 440 unknowns, 384 equations, 356 structural, 28 dependent; 40 docking pins, 84 fixed total
      structural 6.22e-61, dependent 1.94e-61, ||Y||=2.48e+03, alpha=3.24e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=25: 461 unknowns, 400 equations, 372 structural, 28 dependent; 40 docking pins, 89 fixed total
      structural 6.22e-61, dependent 2.24e-61, ||Y||=9.96e+02, alpha=1.23e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=26: 482 unknowns, 416 equations, 388 structural, 28 dependent; 40 docking pins, 94 fixed total
      structural 3.5e-61, dependent 1.94e-61, ||Y||=5.16e+02, alpha=6.88e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=27: 503 unknowns, 432 equations, 404 structural, 28 dependent; 40 docking pins, 99 fixed total
      structural 6.22e-61, dependent 1.56e-61, ||Y||=1.51e+03, alpha=1.89e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=28: 524 unknowns, 448 equations, 420 structural, 28 dependent; 40 docking pins, 104 fixed total
      structural 6.22e-61, dependent 2.48e-61, ||Y||=1.04e+03, alpha=1.31e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=29: 545 unknowns, 464 equations, 436 structural, 28 dependent; 40 docking pins, 109 fixed total
      structural 3.31e-61, dependent 1.56e-61, ||Y||=3.84e+02, alpha=4.77e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=30: 566 unknowns, 480 equations, 452 structural, 28 dependent; 40 docking pins, 114 fixed total
      structural 6.22e-61, dependent 3.69e-61, ||Y||=5.86e+02, alpha=7.43e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=31: 587 unknowns, 496 equations, 468 structural, 28 dependent; 40 docking pins, 119 fixed total
      structural 4.67e-61, dependent 6.51e-61, ||Y||=6.67e+02, alpha=8.49e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=32: 608 unknowns, 512 equations, 484 structural, 28 dependent; 40 docking pins, 124 fixed total
      structural 3.11e-61, dependent 2.14e-61, ||Y||=6.86e+02, alpha=8.71e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=33: 629 unknowns, 528 equations, 500 structural, 28 dependent; 40 docking pins, 129 fixed total
      structural 2.53e-61, dependent 2.33e-61, ||Y||=5.64e+02, alpha=7.14e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=34: 650 unknowns, 544 equations, 516 structural, 28 dependent; 40 docking pins, 134 fixed total
      structural 4.76e-61, dependent 4.08e-61, ||Y||=2.88e+02, alpha=3.66e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=35: 671 unknowns, 560 equations, 532 structural, 28 dependent; 40 docking pins, 139 fixed total
      structural 3.11e-61, dependent 5.25e-61, ||Y||=1.43e+03, alpha=1.77e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=36: 692 unknowns, 576 equations, 548 structural, 28 dependent; 40 docking pins, 144 fixed total
      structural 7.0e-61, dependent 3.11e-61, ||Y||=2.17e+03, alpha=2.79e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=37: 713 unknowns, 592 equations, 564 structural, 28 dependent; 40 docking pins, 149 fixed total
      structural 3.11e-61, dependent 3.4e-61, ||Y||=8.29e+02, alpha=1.06e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=38: 734 unknowns, 608 equations, 580 structural, 28 dependent; 40 docking pins, 154 fixed total
      structural 3.11e-61, dependent 2.14e-61, ||Y||=8.16e+02, alpha=1.01e-10, nullity 8, docking exact True, min|edge| 0.1502 CEF
1=39: 755 unknowns, 624 equations, 596 structural, 28 dependent; 40 docking pins, 159 fixed total
      structural 3.11e-61, dependent 4.67e-61, ||Y||=5.92e+02, alpha=7.70e-11, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=40: 776 unknowns, 640 equations, 612 structural, 28 dependent; 40 docking pins, 164 fixed total
      structural 3.69e-61, dependent 1.94e-61, ||Y||=1.15e+03, alpha=1.43e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=41: 797 unknowns, 656 equations, 628 structural, 28 dependent; 40 docking pins, 169 fixed total
      structural 6.22e-61, dependent 5.45e-61, ||Y||=8.81e+02, alpha=1.11e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=42: 818 unknowns, 672 equations, 644 structural, 28 dependent; 40 docking pins, 174 fixed total
      structural 6.22e-61, dependent 3.11e-61, ||Y||=1.65e+03, alpha=2.12e-10, nullity 8, docking exact True, min|edge| 0.1500 CEF
1=43: 839 unknowns, 688 equations, 660 structural, 28 dependent; 40 docking pins, 179 fixed total

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